5.1 Large Language Models (LLMs)

5.1.1 Concept

- Large Language Models (LLMs) are machine learning models or can be termed as deep learning algorithms, that can understand, comprehend and generate human language text. They work by analyzing massive data sets of language and they are trained using huge data sets. The large amount of data sets makes LLM models capable of understanding and generating natural language and other types of content to perform a wide range of tasks.
- The LLM algorithm converted to a computer program is provided with set of examples, so that the LLM can recognize and interpret human language or other types of complex data. Though the LLMs are trained on data that has been gathered from the Internet and other trusted resources, the output of the data totally depends on the quality of the samples. The quality of the data impacts on how well LLMs will learn natural language, then the developers can exploit the data to the extent that LLM can perform as per expectations.
- LLMs use machine learning techniques named deep learning to understand how characters, words, and sentences function together. Deep learning involves the probabilistic analysis of unstructured data, which eventually enables the deep learning model to recognize distinctions between pieces of content without human intervention.
- Large language models are also referred to as Neural Networks (NNs), which are computing systems inspired by the human brain. These neural networks work using a network of nodes that are layered, much like neurons.
- LLMs are then further trained and fine-tuned, to do the particular task that the programmer wants them to do, such as interpreting and understanding the questions and generating responses, or translating text from one language to another.
- Apart from teaching human languages to Artificial Intelligence (AI) applications, large language models can also be trained to perform a variety of tasks like understanding protein structures, DNA research, writing software code, online searching, Chatbots, customer service and similar tasks for which LLM can be well trained.
- The LLMs work similarly to the human brain. First the human brain is taught, trained and then it becomes capable of doing certain tasks in an expected manner. In the same manner, large language models must be pre-trained and then fine-tuned so that they can solve text classification, question answering, document summarization, and text generation tasks.

5.1.2 Large Language Models Transformer Model

- A **transformer model** is the most common architecture of a large language model. It consists of an encoder and a decoder. A transformer model processes data by tokenizing the input, then simultaneously conducting mathematical equations to discover relationships between tokens.

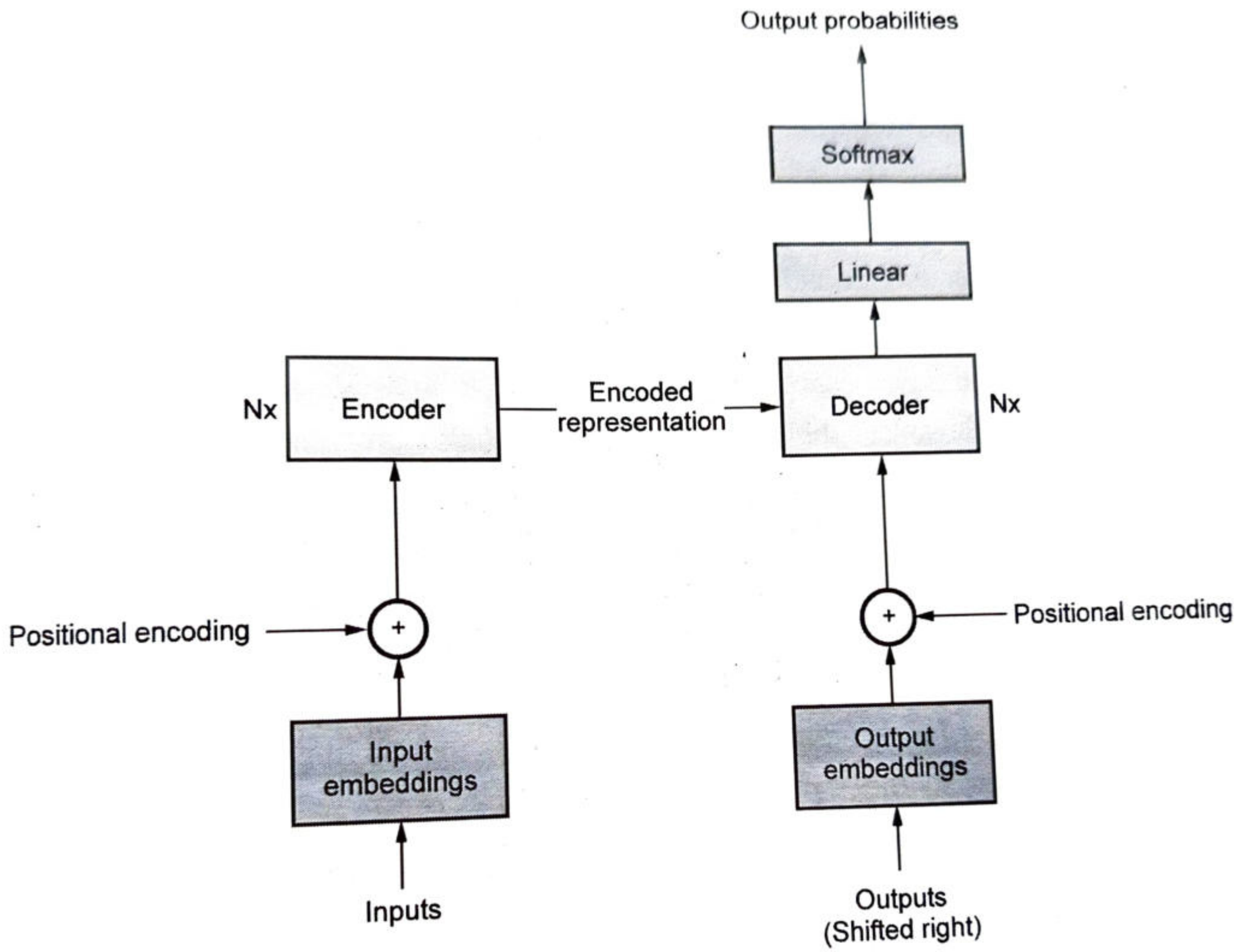

Fig. 5.1.1 Transformer architecture

- The encoder reads and processes the input text, transforming it into a format that the model can understand. Imagine it as absorbing a sentence and breaking it down into its essence. On the other side, the decoder takes this processed information and steps through it to produce the output, like translating the sentence into another language.
- This back-and-forth is what makes transformers so powerful for tasks like translation, where understanding context and generating accurate responses are key. This enables the computer to see the patterns a human would see when the same query is given as input.

5.1.3 Large Language Models Key Components

- Large language models are composed of multiple neural network layers. Various neural networks, such as recurrent layers, feedforward layers, embedding layers, and attention layers work together to process the input text and generate output.
- **The embedding layer** creates embeddings from the input text that captures the semantic and syntactic meaning of the input, so the model can understand the context and relevance.
- **The feedforward layer (FFN)** of a large language model is made up of multiple fully connected layers that transform the input embeddings. While doing this process, these layers enable the model to get insight into higher-level abstractions - helps model to understand the user's intent with the text input.
- **The recurrent layer** interprets the words in the input text in sequence, that captures the relationship between words in a sentence.
- **The attention mechanism** enables a language model to concentrate on single part of the input text that is relevant to the task under consideration. This layer allows the model to generate the most accurate outputs.

5.1.4 Large Languages Model Types

1. **Generic (raw) language model** - This model predicts the next word based on the language in the training data. This language model performs information retrieval tasks.
2. **Instruction-tuned language model** - This model is trained to predict responses to the instructions given in the input. This is useful to perform sentiment analysis, or to generate text or code.
3. **Dialog-tuned language model** - This model is trained to have a dialog by predicting the next response, like the chatbots.

5.1.5 Large Language Models Working

- LLMs operate by leveraging deep learning techniques and vast amounts of textual data. These models are typically based on a transformer architecture, like the generative pre-trained transformer, which excels at handling sequential data like text input.
- A large language model is based on a transformer model and works by receiving an input, encoding it, and then decoding it to produce an output prediction. But before a large language model can receive text input and generate an output prediction, it requires training, so that it can fulfill general functions, and fine-tuning, which enables it to perform specific tasks.

Task 1 - Training - Large language models are pre-trained using large textual datasets from sites like Wikipedia, GitHub, or others. These datasets consist of trillions of words, and their quality will affect the language model's performance. At this stage, the large language model engages in unsupervised learning, meaning it processes the datasets fed to it without specific instructions. During this process, the LLM's AI algorithm can learn the meaning of words, and of the relationships between words. It also learns to distinguish words based on context.

Task 2 - Fine-tuning - In order for a large language model to perform a specific task, such as translation, it must be fine-tuned to that particular activity. Fine-tuning optimizes the performance of specific tasks.

Task 3 - Prompt-tuning - This task is similar to a function to fine-tuning, whereby it trains a model to perform a specific task through few-shot prompting, or zero-shot prompting. A prompt is an instruction given to an LLM. Few-shot prompting teaches the model to predict outputs through the use of examples.

- Over time, deep learning algorithms learn to predict not just the next word that should occur in the sentence, but beyond to the next paragraph and sometimes even the next section. This process is how an LLM bridges the gap between the underlying structure of data and the core business concepts it needs to be able to generate relevant content.
- Whichever is the domain, LLMs must learn the rules of a language and domain-decoding complex patterns to gain a deep understanding of grammar, semantics, and processes so that they can produce contextually precise responses.

5.1.6 Large Language Models Use Cases/Applications

Large language models are useful for business tasks that involve following sub-tasks :

1. Text summarization.
2. Text translation.
3. Text and image generation.
4. Code writing and debugging.
5. Web search enabling search engines to respond to queries, to assisting developers with writing code.
6. Customer service and sentiment analysis.
7. Virtual assistants/chatbots.
8. Text / Document classification.
9. Automated document review and approval.

10. Knowledge base responses to answer queries.
11. Copywriting and technical writing.
12. Large language models have the ability to understand proteins, molecules, DNA, and RNA. This helps to assist in the development of vaccines, finding cures for illnesses.
13. LLMs are used in industries for customer service purposes such as conversational AI, chatbots.
14. Marketing teams can use LLMs to perform sentiment analysis to quickly generate campaign ideas or text as pitching examples.
15. From searching through massive textual datasets to generating legalese, large language models are literally the assistants to lawyers and the entire law community.
16. LLMs can support credit card companies in detecting fraud or similar issues.

5.1.7 Large Language Models Benefits

1. **Efficiency** - LLMs can significantly improve the efficiency of processes due to their ability to understand and process natural language at a large scale with huge data processing.
2. **Large set of applications** - LLMs can be used for language translation, sentence completion, sentiment analysis, question answering, mathematical equations, and such tasks.
3. **Cost reduction** - With LLMs, tasks such as customer support, data analysis, and others can be automated, thus reducing operational costs.
4. **Continuous improvement** - Large language model performance is continually improving because it grows when more data and parameters are added. In other words, the more it learns, the better it gets. Large language models can exhibit what is called "**in-context learning**."
5. **Data analysis** - LLMs can analyze and interpret vast amounts of data faster and more effectively than humanly possible, providing businesses with valuable insights quickly.
6. **Real time response to customer** - LLM-based applications can enhance customer interactions by offering personalized assistance and real-time responses within stipulated time.
7. **Scalability** - LLMs can handle an increasing amount of work due to their deep learning process.

5.1.8 Large Language Models Limitations

LLMs have various limitations and challenges that they faced :

1. **Hallucinations** - A hallucination is event when a LLM produces an output that is false, or that does not match the user's intent. The result can sometimes be what is referred to as a "hallucination."
2. **Security** - LLMs face important security risks when not managed or surveilled properly. They can leak people's private information, participate in phishing scams, and produce spam.
3. **Ethical concerns and bias** - LLMs are trained on vast amounts of data from many sources, so they might reflect and reproduce the biases present in those data sets. The data used to train language models will affect the outputs a given model produces. If the data represents a single demographic, or lacks diversity, the outputs produced by the large language model will also lack diversity and hence bias will be inherent.
4. **Consent** - LLMs are trained on trillions of datasets - some of which might not have been obtained consensually or unknown to owners. When scraping data from the internet, large language models have been known to ignore copyright licenses, plagiarize written content, and repurpose proprietary content without getting permission from the original owners or artists.
5. **Scaling** - It is very critical to scale on time- and resource-consuming factors so as to maintain large language models.
6. **Deployment** - Deploying large language models requires deep learning, a transformer model, distributed software and hardware, and overall technical expertise.

5.2 Use-cases - ChatGPT, Gemini, Bhashini, Krutrim

5.2.1 ChatGPT

5.2.1.1 Introduction

- Chat GPT stands for Chat Generative Pre-Trained Transformer and was developed by an AI research company, Open AI.
- OpenAI LP, a capped-profit research and deployment company, governed by the OpenAI non-profit board, released ChatGPT on November 30, 2022. OpenAI is an artificial intelligence research laboratory founded by Elon Musk, Sam Altman, Greg Brockman, Ilya Sutskever, Wojciech Zaremba, and John Schulman. The venture is supported by investors including Microsoft, Reid Hoffman's charitable foundation, and Khosla Ventures.

- It is an Artificial Intelligence (AI) chatbot technology that can process natural human language and generate a response.
- The language model can respond to questions and compose various written content, including articles, social media posts, essays, code and emails.
- It can be referred to as powerful text-generating dialogue system.
- **ChatGPT is free to use**, and Free tier users now have access to a large range of capabilities with GPT-4o, including access to a series of tools and access to GPTs. The free version of ChatGPT is available to everyone. Paid plans (Plus, Team, and Enterprise) are priced per user per month.

5.2.1.2 Features and Use-cases

- ChatGPT has ability to generate responses.
- Solve math equations.
- Translate from one language to another.
- Debug and fix code.
- Write a story/poem.
- Classify things.
- Script social media posts.
- Summarize articles, podcasts or presentations.
- Research markets for products.
- Reword existing content for a different medium, such as a presentation transcript for a blog post.
- Explain complex topics.
- ChatGPT can help building virtual assistants for help with scheduling email management, and customer service.
- ChatGPT can be used to generate automated responses to common customer inquiries and forward them to the customer.
- ChatGPT can be used to build a knowledge base that can answer common questions and provide information to users.
- ChatGPT can be used to provide automated assistance to users who need help with IT-related issues, such as password resets and account lockouts, in general the IT help desk management.
- ChatGPT can be used to provide automated assistance to employees who need help with HR-related issues, such as benefits and time-off requests.

- Simple user interface and hence easy to use.

5.2.1.3 Working

ChatGPT Working Phases

- ChatGPT uses machine learning algorithms to process and analyze large amounts of data to generate responses to user inquiries. ChatGPT works through its Generative Pre-trained Transformer, which uses specialized algorithms to find patterns within data sequences.
- It fetches from a large dataset of text, ChatGPT uses this dataset to learn about language, grammar, and the structure and meaning of words and sentences.
- This enables it to understand the context and intent of user queries and generate appropriate responses.
- ChatGPT works on a neural network-based architecture called a **transformer**. Datasets from websites, books, and articles are used for training the model on language patterns and structure.
- It learns to predict the next word based on the previous one. Once trained, the model can generate new text by predicting the next word in a sentence, with a given prompt or context. The process is reiterated until the model has generated a complete sentence or the required number of words.
- The model also uses an attention mechanism while generating text. The mechanism allows selective focus on certain parts of the input for more accurate and coherent responses.
- When using ChatGPT for conversational AI, the model is typically fine-tuned on a smaller dataset of conversational text to further improve its ability to generate human-like responses.
- ChatGPT originally used the GPT-3 large language model, a neural network machine learning model and the third generation of generative pre-trained transformer. The transformer pulls from a significant amount of data to formulate a response.

ChatGPT Training

- ChatGPT uses reinforcement learning training begins with generic data, then moves to more tailored data for a specific task. ChatGPT was trained with online text to learn the human language, and then it used transcripts to learn the basics of conversations.
- Human trainers provide conversations and rank the responses. These reward models help determine the best answers. To keep training the chatbot, users can up-vote or down-vote its response by clicking on thumbs-up or thumbs-down icons beside the answer. Users can also provide additional written feedback to improve and fine-tune future dialogue.

5.2.1.4 Limitations and Disadvantages

- It does not understand human language completely.
- It cannot understand sarcasm and irony.
- It may occasionally generate incorrect information.
- Responses may have a feel of machine generation.
- It may occasionally produce harmful instructions or biased content.
- GPT based models like ChatGPT are good at generating human-like text, but they lack understanding of the context. Therefore, it may generate irrelevant or nonsensical responses in certain situations.
- It has limited knowledge of the world and events after 2021, as the further data is yet to be fed. Which means one will not get answers to questions such as, "What is the weather like today". Also, for specific customised, enterprise **(DWP)** training it will need huge amount of data for reinforced learning and supervised training which will need continuous development effort.
- ChatGPT uses text based on input, so it could potentially reveal sensitive information. The model's output can also track and profile individuals by collecting information from a prompt and associating this information with the user's phone number and email. The information is then stored indefinitely.
- Using ChatGPT in an enterprise service may have legal and ethical implications, such as potential violations of privacy and data protection laws, or issues related to consent and transparency.
- Being an open source model, support specifically for enterprise customer environments will be an issue.
- GPT-based models like ChatGPT are computationally expensive, they require powerful hardware and infrastructure to run, and fine-tuning a model may also require a large amount of computational resources.

5.2.2 Gemini

5.2.2.1 Introduction

- The Gemini app is **an AI assistant** that gives access to large language models. Gemini was earlier known as Bard.
- The language models learn by 'reading (taking input)' trillions of words that help them pick up on patterns that make up language. This allows them to pick up on and reply to your questions using common language patterns.

- Gemini is always in learning mode, which means that applications learn from user prompts, responses and feedback. Gemini will make mistakes and might even say something offensive or absurd.
- Google's Gemini AI offers two main options : **A free version and a paid upgrade called Gemini Advanced**. Both use powerful artificial intelligence to help with tasks and answer questions. The free version works well for basic needs, while the paid version offers more advanced features.

5.2.2.2 Features and Use-cases

- Gemini AI has ability to reason and explain.
- Gemini AI is designed to answer the question with reason and explanation. If we compare Gemini and search engine which only shares links in response to a question, Gemini provides explanation. In other words, it turns a complex task into a conversation by pulling in information from several sources.
- Gemini AI is designed to be multimodal, meaning that the AI chatbot can understand questions by text, photo, and even video. For instance a user can take a photo of a holiday packing list and the location user is heading to and ask Gemini AI if anything is missing.
- Gemini connects with Google's other apps, where information can not only be pulled from other apps but also exported to Google's other applications like Maps or Google Docs.
- As Gemini connects with other Google services, it can read associated user emails. Gemini can converse with user based on that email. So, one can ask specific questions about your mail or summarise the most important emails from inbox.
- Gemini can parse complex visuals, such as charts, figures and diagrams, without external OCR tools. It can be used for image captioning and visual Q&A capabilities.
- Gemini AI has ability to identify products and objects. One can launch Gemini on Android or iOS, click on the photo button, snap a photo and then ask Gemini to identify a product or object in the photo. For example, Gemini can identify a beverage and it will not only identify the drink but also provide ingredients with tiny details.
- Gemini AI can do coding tasks including the ability to translate code between languages. Users can use Google's AI assistant to convert code written in one language to another. It can also be used to generate different coding solutions for the same problem or fill in the missing parts of a code or debug existing errors.
- Gemini has support for **speech recognition** across more than 100 languages and audio translation tasks.

- Gemini AI is adaptable to different applications and systems. Google developed Gemini as a foundation model to be widely integrated across various Google services and made available for developers to use in building their own applications. Applications that use Gemini include.
  - **AlphaCode 2 - DeepMind's code** generation tool makes use of a customized version of Gemini Pro.
  - **Vertex AI** - Google Cloud's Vertex AI service, which provides foundation models that developers can use to build applications that can access Gemini Pro.
  - **Search** - Google is experimenting with using Gemini in its search generative experience to reduce latency and improve quality.
  - **Google Pixel** - The Google-built Pixel 8 Pro smartphone is the first device engineered to run Gemini Nano, where Gemini powers new features in existing Google apps, such as summarization in Recorder and Smart Reply in Gboard for messaging apps.
  - **Android 14** - The Pixel 8 Pro is the first Android smartphone to benefit from Gemini. Android developers can build with Gemini Nano through the AICore system capability.
  - **Google AI Studio** - Developers can build prototypes and apps with Gemini using the Google AI Studio web-based tool.

5.2.2.3 Working

- Gemini uses natural language processing and machine learning. Gemini integrates NLP capabilities, which provide the ability to understand and process language.
- Google Gemini works by first being trained on a massive corpus of data. After training, the model uses several neural network techniques to be able to understand content, answer questions, generate text and produce outputs.
- Gemini LLMs use a transformer model-based neural network architecture. The Gemini architecture has been enhanced to process lengthy contextual sequences across different data types, including text, audio and video. Google DeepMind makes use of efficient attention mechanisms in the transformer decoder to help the models process long contexts, spanning different modalities.
- Gemini models have been trained on diverse multimodal and multilingual data sets of text, images, audio and video with Google DeepMind using advanced data filtering to optimize training. As different Gemini models are deployed in support of specific Google services, there's a process of targeted fine-tuning that can be used to further optimize a model for a use case.

- During both the training and inference phases, Gemini benefits from the use of Google's latest tensor processing unit chips, TPU v5, which are optimized custom AI accelerators designed to efficiently train and deploy large models as per the requirements.

5.2.2.4 Limitations and Disadvantages

- Gemini must learn to give correct answers. To do this, the models must be trained on correct information that's not inaccurate or misleading. However, they also must be able to identify incorrect or misleading information when it comes their way.
- Gemini can be used for everything from multimodal storytelling to personalised search, but the fact is that it is far from perfect. There were times when AI summaries from Gemini recommended users eat rocks and put glue on their pizza, and unlike ChatGPT, there are no footnotes with links to where the answers are sourced from.
- As there's always new information to learn, AI training is an endless, compute-intensive process. Across all Gemini models, Google has claimed it has followed responsible development practices, including extensive evaluation to help limit the risk of bias and potential harm.
- There are limits on how original and creative the content Gemini produces. This is particularly the case with the free version, which has trouble processing complicated prompts, with multiple steps and nuances, and producing adequate output.

5.2.3 Bhashini

5.2.3.1 Introduction

- Bhashini application is **developed to help Indian citizens translate content in different Indian languages.** The program's main objective is to develop services and products for citizens by leveraging the power of artificial intelligence and other emerging technologies. Bhashini is an AI-based translation tool which is designed to break the barrier between the diverse languages that people speak across India.
- **Bhashini**, developed by MeitY Government of India, is a free **Android** app that aims to build a national public digital platform for languages.
- **Bhashini** is a unique platform that aims to provide Indian citizens with a comprehensive and user-friendly interface to access various language services in a single place.
- Bhashini is a free to use application.

5.2.3.2 Features and Use-cases

- **Bhashini**, can be used to access a wide range of language services such as translation, transliteration, text-to-speech, speech-to-text, and much more.
- It is long-term **STRATEGY** for developing sustainable Indian language technology, solutions and ecosystem technology.
- Many Indian languages, such as Hindi, Bengali, Tamil, Telugu, Marathi, Gujarati, Kannada, Malayalam, Punjabi, and others, are supported by Bhashini. Its capacity to be multilingual is essential since it allows it to accommodate a variety of language preferences.
- The app is designed to support various Indian languages, including Hindi, Punjabi, Tamil, Telugu, Kannada, Malayalam, and Bengali. It **seeks to enable easy access to the internet and digital services in Indian languages**, including voice-based access, and **help the creation of content in Indian languages**.
- The app's user-friendly interface and the wide range of language services make it a must-have app for anyone who needs language services on-the-go.
- By utilizing sophisticated deep learning architectures, Bhashini is able to understand the subtleties of context in Indian languages, taking into account grammatical structures, idiomatic phrases, and linguistic nuances unique to each language.
- Bhashini aims **to make Artificial Intelligence and Natural Language Processing** (NLP) **resources available in the public domain** to be used by -- Indian startups and individual innovators.
- Bhashini **helps developers to offer all Indians easy access to the internet and digital services in their native languages.**
- Due to the model's versatility across several domains - including healthcare, finance, education, and more - it may be tailored to meet business demands and improve language processing in specialized fields.
- Bhashini has **a separate 'Bhasadaan' section which allows individuals to contribute to multiple crowdsourcing initiatives,** and it is also accessible via respective Android and iOS applications.

5.2.3.3 Working

- Bhashini's technological architecture is based on deep-learning techniques specifically designed for the intricacies of Indian languages.
- Bhashini uses transformer-based neural networks, taking cues from well-known designs such as BERT, GPT, or XLNet. Bhashini's design, in contrast to generic models, is carefully tailored and adjusted to fit the nuances of Indian languages, guaranteeing its

effectiveness in a variety of linguistic contexts.

- The training data, a sizable corpus of varied text sources including books, news articles, social media, and numerous digital platforms in several Indian languages, is what makes the model strong. Bhashini's training is based on this large dataset, which helps it understand the subtleties of language usage, idiomatic phrases, grammatical structures, and contextual changes unique to each language and its style.
- Bhashini uses particular tokenization and preprocessing methods to manage the variety of scripts and character sets present in Indian languages. These methods enable effective processing and representation inside the model by segmenting the input text into tokens or subword units specific to each language.
- The training phase is critical where the model adjusts to the linguistic nuances and subtleties found in various Indian languages. It entails painstaking fine-tuning and optimization.
- Bhashini's performance and adaptability are improved by methods of domain-specific fine-tuning and transfer learning, which guarantee its relevance and application in specialized industries.
- Bhashini's technological architecture combines state-of-the-art deep learning principles with customized data pretreatment and optimization approaches to produce a model that can analyze and generate text across India's heterogeneous linguistic environment with effectiveness.

5.2.3.4 Limitations and Disadvantages

- The availability of high-quality and diverse training data across all Indian languages remains a challenge, impacting the model's performance and coverage.
- Training and maintaining language models like Bhashini require significant computational resources and expertise, posing a barrier to scalability.

5.2.4 Krutrim

5.2.4.1 Introduction

- **Krutrim AI is an LLM that understands and generates content in 10 Indian languages.** The aim is to be able, through the technology of a Large Language Model (LLM), that reflects India's cultural and linguistic diversity but also **outperforms all other LLMs like ChatGPT and GPT-4 on benchmarks for Indian languages.**
- Indian multinational ridesharing company, Ola, introduced Krutrim AI, claiming it as "India's own AI".

- Krutrim means artificial in Sanskrit and comes in base and pro models. Like ChatGPT, this primary version can respond to prompts or queries from the public. It can comprehend around 22 Indian languages and generate text in 10 languages, including Marathi, Hindi, Bengali, Tamil, Kannada, Telugu, Odia, Gujarati, and Malayalam
- Krutrim has two versions namely, the basic Krutrim model and another more powerful version - **Krutrim Pro**.
- The model aims to empower each of the 1.4 billion+ Indian consumers, developers, entrepreneurs, and enterprises by giving them the power of AI.

5.2.4.2 Features and Use-cases

- **Krutrim AI understands both textual and voice inputs in 10 Indian languages, such as Hindi, Kannada, Marathi, Odiya and Telugu.** It also produces outputs of the same type. It therefore can serve a big and eclectic user base, in particular those who are uncomfortable with English or other foreign languages.
- Krutrim can switch between languages.
- Because **Krutrim AI has been trained on over 2 trillion tokens collected from local sources - twenty times more than any other model** - it can deal much better with the language characteristics of each and every one used. In terms of Indic language support, it has an edge over other LLMs.
- Krutrim uses a custom tokenizer for interpreting the languages and scripts, Krutrim AI is also capable of processing and generating content in different linguistic settings.
- **Krutrim AI can do things such as vision, speech and task execution.** This means that in addition to recognizing and outputting text and speech, Krutrim AI can also recognize images or grasp actions.
- **Krutrim AI can deal with complicated and exclusive inquiries about content in all kinds of industries.**
- Krutrim AI says that it observes responsible use of artificial intelligence and protects user privacy and data security.
- Krutrim can do content Generation and Translation. Leveraging natural language generation and multimodal techniques, Krutrim AI can be used to develop creative content of various types in 10 Indian languages. **This includes the creation of poems, stories, folk songs, jokes or memes in whatever genre and style user preferred.**
- The Krutrim AI allows for the processing and generation of speech in 10 Indian languages, thus making possible voice applications and recognition systems.

5.2.4.3 Working

- Krutrim architecture employs a neural network capable of learning the relationships between words and sentences so that it can create coherent, relevant text.
- **Krutrim AI has 1.5 billion parameters, which are the weights that affect how a model processes inputs and creates outputs.** Krutrim AI has been trained on several **GPUs** and TPUs, which are processed hardware devices capable of performing complex mathematical operations at high speed.
- Krutrim AI has been assessed on several industry-standard benchmarks including GLUE, XGLUE and IndicGLUE. **These measure capability in Natural Language Processing (NLP) as well as understanding tasks like sentiment analysis, question answering or machine translation of text** from one human language into another.
- Training is most crucial phase in the working of Krutrim, that is trained in **20 Indian languages, that has been trained on more than 2 trillion tokens gathered through sources located in India.** These are sub-words used in conversation and text. Specializing in interpretation of languages and scripts, Krutrim AI uses a custom tokenizer to split the data into smaller units which can be fed as input arguments for the model.
- Krutrim AI can be applied to many different domains and platforms, including chatbots, content generation speech applications, semantic search or creative content.
- **Krutrim Basic** version is open to public use, which is **free of charge for personal and non-commercial use, but only up to 10,000 tokens a month**. Users can register on the Krutrim website to use the base model.
- **Krutrim Pro**, the more powerful and multi-modal variant of Krutrim AI, which will be introduced for enterprise use in next quarter's issue. **Krutrim Pro costs up to 1000 INR per month for up to 10 users; extra tokens and features will be charged additionally.** To be able to try out the pro model early, companies and developers can register on the Krutrim website.

5.2.4.4 Limitations and Disadvantages

- Any AI model will have a hallucination problem, as it is a probabilistic model. Same is the issue with Krutrim, sometimes it may respond weirdly. Training is a continuous process that it needs to work efficiently.
- Krutrim needs authentic, relevant and factual Indian open web data to train its models, which is critical task.

5.3 Current Issues and Future Challenges of AI

1. Rise of ChatGPT has brought large language models in the market and activated speculation and questions raised on what the future might look like.
2. As large language models continue to grow and improve their command of natural language, there is much concern regarding what their advancement would do to the job market. It's clear that large language models will develop the ability to replace workers in certain fields.
3. Scaling up the volume of pre-training data has been a key driver in enhancing the capabilities of Large Language Models (LLMs) to handle a wide range of tasks.
4. Tokenization refers to the process of dividing a sequence of text into smaller units called tokens, which are then fed into a model. Tokenizers introduce challenges like :
   a. Computational overhead
   b. Language dependence
   c. Vocabulary size limitations
   d. Information loss
   e. Reduced human interpretability.

   Subword-level inputs are prevalent, striking a balance between vocabulary size and sequence length.
5. Fine-tuning is a technique used to adapt pre-trained LLMs to smaller, domain-specific datasets, enhancing their performance for particular tasks. This approach involves further training the model using either the original language modeling objective or by adding learnable layers to the pre-trained model's output representations to align them with specific downstream tasks. However, fine-tuning entire LLMs presents challenges due to large memory requirements, making it impractical for many practitioners. Storing and loading individual fine-tuned models for each task incurs computational inefficiency.
6. Factual information acquired during the initial training phase of a model may contain errors or become outdated over time, such as failing to account for political leadership changes. However, retraining the model with new data is costly, and attempting to replace old facts with new ones during fine-tuning is challenging.
7. There are two primary reasons for the high inference latencies observed in Large Language Models (LLMs). First, the inference process of LLMs is hindered by low parallelizability, as it operates token by token. Second, the considerable memory footprint of LLMs is attributed to the model's size and the transient states necessary during decoding, such as attention key and value tensors.

8. LLMs often suffer from hallucinations, wherein they generate linguistically coherent text that contains inaccurate information. These hallucinations can be difficult to identify due to the fluency of the text. The source content provided to the model, such as prompts or context, is considered to categorize hallucinations. Intrinsic hallucinations involve text that contradicts the source content, while extrinsic hallucinations are generated outputs that cannot be verified or contradicted by the source.
9. In today's fast-paced digital world, Large Language Models (LLMs) are changing how businesses work with data. However, it's not easy to make one from scratch. Though these AI systems are advanced, making your own model comes with its own set of problems that not many people really understand.
10. Custom AI solutions face significant challenges in managing the complexity of these models.
11. Exercising control over the data utilized for training ensures that sensitive or proprietary information remains confined within the user organization.
12. When in the right hands, large language models are used efficiently with the ability to increase productivity and process efficiency, but this has posed ethical questions for its use in the human world.

Review Questions

1. *Write a note on below topic giving suitable application*
   a. *Large language models* **(Refer section 5.1)**
   b. *ChatGPT* **(Refer section 5.2.1)**
   c. *Gemini* **(Refer section 5.2.2)**
   d. *Bhashini* **(Refer section 5.2.3)**
   e. *Krutrim* **(Refer section 5.2.4)**